

Anchored Physical Model of the Coupled Levitated-Sensor Array: Equations, Optimization Results, and Open Parameters

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Abstract

This note documents the physical (SI) sensitivity pipeline of **ResOSc** after anchoring it to the published LSD parameters [PRL **128**, 111101 (2022), Table I] and to the levitated-sensor proposal of Arvanitaki & Geraci [PRL **110**, 071105 (2013), “AG13”]. Every equation used in the pipeline is stated with all terms defined (Sec. 1); parameter provenance is tabulated (Sec. 2); the Metropolis optimization and the three-way comparison results are summarized (Secs. 3–4). Headline: with the anchored model, the cavity readout (A) is thermal-noise-limited and the optimal array is the uniform stack at the top of the trap range (three-way tie, $d_{\max} = 5.5$ pc at $\mathcal{M}_c = 10^{-3}M_{\odot}$); the imaging readout (B) is shot-noise-limited and there *Coulomb coupling wins*: the optimized coupled array reaches $2.2\times$ the best uncoupled array and $12\times$ the as-built baseline, with its deepest sensitivity buckets on the thermal floor inside the LSD target band. Absolute forecasts currently hinge on a small set of unconfirmed parameters and conventions collected as direct questions in Sec. 5.

1 The sensitivity chain, with every term defined

We consider N optically levitated discs in a cavity of length L , with positions $x_i(t)$ measured along the cavity axis.

1.1 Mechanics and normal modes

The equations of motion are

$$M\ddot{\mathbf{x}} + (\text{damping}) + K\mathbf{x} = \mathbf{F}(t), \quad (1)$$

where $\mathbf{x} = (x_1, \dots, x_N)^T$ are the axial displacements [m], $M = \text{diag}(m_1, \dots, m_N)$ the masses [kg], $\mathbf{F}(t)$ the external forces [N], and K [N/m] the stiffness matrix assembled from the optical trap stiffnesses $k_{\text{trap},i} = m_i(2\pi f_{\text{trap},i})^2$ ($f_{\text{trap},i}$ = trap frequency [Hz]) on the diagonal plus the inter-particle couplings k_{ij} in the standard spring (graph-Laplacian) pattern: $K_{ii} = k_{\text{trap},i} + \sum_{j \neq i} k_{ij}$, $K_{ij} = -k_{ij}$. The couplings come from Coulomb interaction between charges q_i (units of the elementary charge e) at equilibrium separations d_{ij} :

$$k_{ij} = \frac{2 q_i q_j e^2}{4\pi\epsilon_0 d_{ij}^3}, \quad (2)$$

with ϵ_0 the vacuum permittivity; all pairs are included ($1/d^3$ tails), and equal-sign charges give $k_{ij} > 0$ (repulsive longitudinal spring). Adjacent-disc spacing: $d = 15.5 \mu\text{m}$ (companion SNR note).

The generalized eigenproblem $K\mathbf{v} = \omega^2 M\mathbf{v}$ yields normal modes $n = 1..N$ with frequencies $f_n = \omega_n/2\pi$ [Hz] and unit-norm shapes $\mathbf{v}_n$ (columns of V). Modal masses and the two modal damping rates (see Sec. 1.3 for the distinction) are mass-weighted averages over the shape:

$$\mu_n = \mathbf{v}_n^T M \mathbf{v}_n \text{ [kg]}, \quad \Gamma_n = \frac{\sum_i V_{in}^2 m_i \gamma_i}{\mu_n}, \quad \Gamma_n^f = \frac{\sum_i V_{in}^2 m_i \gamma_i^f}{\mu_n} \text{ [Hz]}, \quad (3)$$

where γ_i is the per-disc mechanical *linewidth* and γ_i^f the per-disc *force-noise* damping, both in Hz (cycles/s; angular rates are $2\pi\times$ these). Each mode responds to force at frequency f through the susceptibility

$$\chi_n(f) = \frac{1}{\mu_n[(2\pi f_n)^2 - (2\pi f)^2 + i(2\pi f)(2\pi \Gamma_n)]} \text{ [m/N]}. \quad (4)$$

1.2 Gravitational-wave drive (AG13, exact)

A GW of strain $h(t)$ (dimensionless), propagating perpendicular to the cavity, drives disc i with the inertial force [AG13 Eq. (5)]

$$f_i^{\text{sig}}(t) = \frac{1}{2} m_i (2\pi f_{\text{gw}})^2 (L - x_i) h(t) \approx \frac{1}{2} m_i (2\pi f_{\text{gw}})^2 L h(t), \quad (5)$$

where f_{gw} is the *wave's* frequency, L the *full* cavity length, and $x_i \ll L$ (discs near the input mirror). Physically: the trap minima are locked to the intracavity interference pattern, which breathes with the wave by $\frac{1}{2}hL$; the disc's inertia resists, giving a force = mass $\times$ reference acceleration. Equation (5) is exact at all frequencies and gives free-mass (flat, LIGO-like) displacement response $\frac{1}{2}hL$ above resonance. The companion note's Eq. (20), $\beta_i = k_{\text{trap},i}L$, is the quasi-static ($f_{\text{gw}} \rightarrow$ resonance) limit *with L the half-baseline*; the two conventions differ by a factor 2 in L (Question 6).

The modal drive and the strain-to-observable transfer function are

$$B_n(f) = \frac{1}{2}(2\pi f)^2 L (\mathbf{v}_n^T \mathbf{m}) \text{ [N/strain]}, \quad T(f) = \sum_n N_n B_n(f) \chi_n(f) \text{ [m/strain]}, \quad (6)$$

where $\mathbf{m} = (m_1, \dots, m_N)^T$ and $N_n = \mathbf{v}_n^T \mathbf{w}$ is the modal projection of the readout weight vector $\mathbf{w}$ (the observable is the single time series $O(t) = \sum_i w_i x_i(t)$).

1.3 Thermal noise and the damping split

Two distinct damping rates enter (this is the key correction from the anchoring):

- **Force-noise damping** γ_i^f : real bath friction, obeying the fluctuation–dissipation theorem (FDT). It is the sum of residual-gas damping and the photon-recoil-heating equivalent,

$$\gamma_i^f = \gamma_g + \frac{\hbar(2\pi f_{\text{trap},i})}{k_B T} \gamma_{\text{sc},i}, \quad \gamma_g = \frac{\hbar(2\pi f_0)}{k_B T} (N_i \gamma_g)_{\text{pub}} \approx 2.7 \times 10^{-9} \text{ Hz}, \quad (7)$$

where T is the gas temperature [K], k_B Boltzmann's constant, $\hbar$ the reduced Planck constant, $\gamma_{\text{sc},i}$ the photon-recoil rate (0.05 Hz at $f_{\text{trap}} = 100$ kHz, scaling $\propto f_{\text{trap}}$; PRL 128 Table I), and $(N_i \gamma_g)_{\text{pub}} = 0.17$ Hz at 100 kHz is the *published product* of the mean phonon occupation $N_i = k_B T / \hbar \omega_0 \approx 6.3 \times 10^7$ and the gas damping. N.B. the published 0.17 Hz is *not* the mechanical linewidth (Question 1).

- **Mechanical linewidth** $\gamma_i = f_{\text{trap},i}/Q_{\text{eff},i}$: set by feedback cooling (cold damping), which adds damping *without* force noise in the idealized limit. $Q_{\text{eff},i}$ is a per-disc control-loop setting (AG13 uses 5.4×10^5 ; tunable).

The one-sided modal thermal force PSD (power spectral density) and its appearance at the observable are

$$S_{Q_n} = 4k_B T \mu_n (2\pi\Gamma_n^f) \text{ [N}^2/\text{Hz]}, \quad S_O^{\text{th}}(f) = \sum_n N_n^2 S_{Q_n} |\chi_n(f)|^2 \text{ [m}^2/\text{Hz}]. \quad (8)$$

Because thermal noise is a force, it is filtered by the same $|\chi_n|^2$ as the signal; on resonance the thermal-limited strain noise is therefore linewidth- and weight-independent.

1.4 Readout (shot) noise

Readout A (cavity probe; weights hardwired). All discs dispersively shift one cavity mode with per-disc coupling

$$G = k_c \frac{V_d}{4V_c} (\epsilon - 1) \omega_c \text{ [rad s}^{-1}\text{m}^{-1}], \quad V_c = \frac{\pi w_0^2 L}{4}, \quad (9)$$

with $k_c = 2\pi/\lambda$ the probe wavevector, $\omega_c = 2\pi c/\lambda$ its angular frequency, $\lambda = 1.55 \mu\text{m}$, V_d the disc (stack) volume, ϵ its dielectric constant, w_0 the cavity waist, and c the speed of light. The measured phase carries $\sum_i G_i x_i$, fixing the weights $\mathbf{w} = \mathbf{G}/|\mathbf{G}|$. The phase shot noise $S_\phi = \hbar\omega_c/(\eta P_{\text{det}})$ [rad²/Hz] (η = detection quantum efficiency, P_{det} = detected power) gives the displacement-equivalent readout noise

$$S_{O,A}^{\text{ro}}(f) = \frac{S_\phi}{\sum_i G_i^2} \left(\frac{\kappa}{2}\right)^2 \left[1 + \left(\frac{f}{f_{\text{cav}}}\right)^2\right], \quad \kappa = \frac{\pi c}{\mathcal{F}L}, \quad f_{\text{cav}} = \frac{\kappa}{2\pi}, \quad (10)$$

with κ the cavity decay rate [rad/s] and $\mathcal{F}$ the cavity finesse. This matches AG13's $\sqrt{S_z} = \frac{\kappa}{4k_c g} \sqrt{\hbar\omega_c/P_d} \sqrt{1 + 4\omega^2/\kappa^2}$ up to factor-2 detection conventions (verified against their quoted $3.8 \times 10^{-16} \text{ m}/\sqrt{\text{Hz}}$).

Readout B (per-disc imaging; weights free). Each disc is imaged independently with per-disc position imprecision

$$S_i^{\text{ro}} = \frac{\hbar \omega_{\text{ro}} \lambda_{\text{ro}}^2}{4\pi \eta \text{NA}^2 P_i^{\text{sc}}} \text{ [m}^2/\text{Hz]}, \quad S_{O,B}^{\text{ro}} = \sum_i w_i^2 S_i^{\text{ro}}, \quad (11)$$

where λ_{ro} ($\omega_{\text{ro}} = 2\pi c/\lambda_{\text{ro}}$) is the imaging wavelength, NA the collection numerical aperture, and P_i^{sc} the collected scattered power. The weights $\mathbf{w}$ are a free (unit-norm) post-processing choice; frequency-dependent weights $\mathbf{w}(f)$ are a strict further improvement not yet implemented.

1.5 Strain-referred noise, source, and figure of merit

$$S_h(f) = \frac{S_O^{\text{th}}(f) + S_O^{\text{ro}}(f)}{|T(f)|^2} \text{ [Hz}^{-1}] \quad (12)$$

is the strain-referred noise PSD; $\sqrt{S_h}$ is the sensitivity curve. The source is an equal-mass primordial-black-hole binary of chirp mass $\mathcal{M}_c = (m_1 m_2)^{3/5}/(m_1 + m_2)^{1/5}$ with the stationary-phase inspiral spectrum (Maggiore)

$$|\tilde{h}(f)|^2 = \mathcal{A}^2 f^{-7/3}, \quad \mathcal{A}^2 = \frac{5}{24} \frac{c^2}{\pi^{4/3} r^2} \left(\frac{G\mathcal{M}_c}{c^3}\right)^{5/3} \Theta, \quad (13)$$

with r the source distance, G Newton’s constant, and Θ an $O(1)$ orientation factor (set to 1). The spectrum ends at the innermost-stable-circular-orbit frequency $f_{\text{ISCO}} = c^3/(6^{3/2}\pi GM_{\text{tot}})$ with $M_{\text{tot}} = 2^{6/5}\mathcal{M}_c$ for equal masses (≈ 1.9 MHz at $\mathcal{M}_c = 10^{-3}M_\odot$: the entire 10–300 kHz band is live). The matched-filter signal-to-noise ratio and horizon distance are

$$\rho^2 = 4 \int_{f_{\text{lo}}}^{f_{\text{hi}}} \frac{|\tilde{h}(f)|^2}{S_h(f)} df, \quad d_{\text{max}} = r_{\text{ref}} \frac{\rho(r_{\text{ref}})}{\rho_{\text{thr}}}, \quad \rho_{\text{thr}} = 8, \quad (14)$$

using $\rho \propto 1/r$. Band: 10–300 kHz; benchmark $\mathcal{M}_c = 10^{-3}M_\odot$ ($\approx$ one Jupiter mass; middle of the PRL 133, 111401 search range).

2 Parameter anchoring and validation

Table 1: Benchmark parameters and provenance. “128” = PRL 128, 111101 Table I; “AG13” = PRL 110, 071105; “note” = companion SNR note; **TBC** = unconfirmed.

Parameter	Symbol	Value	Source
Trap frequency range	f_{trap}	10–100 kHz	128
Cavity length	L	10 m	128
Pressure / temperature	P, T	10^{-11} Torr, 300 K	128
Occupation $\times$ gas damping	$N_i \gamma_g$	0.17 Hz (100 kHz)	128 (Q1)
Photon-recoil rate	γ_{sc}	0.05 Hz (100 kHz), $\propto f$	128 (Q1)
Derived gas damping	γ_g	2.7×10^{-9} Hz	derived
Effective Q (feedback)	Q_{eff}	5.4×10^5 baseline; cap TBC	AG13 (Q2)
Disc stack mass	m	5.8×10^{-10} kg	geometry $\checkmark$
Stack volume, $\epsilon - 1$	$V_d, \epsilon - 1$	2.6×10^{-13} m ³ , 1.1	geometry
Probe wavelength, waist	λ, w_0	1.55 μm , 75 μm	128/AG13
Cavity finesse	$\mathcal{F}$	10 ($f_{\text{cav}} = 1.5$ MHz)	AG13
Detection power (A)	P_{det}	0.2 mW	AG13
Imaging aperture (B)	NA	0.5 (“up to”; was 0.1)	Nancy 8/31 (Q3)
Imaging wavelength (B)	λ_{ro}	532 nm	note, TBC
Collected power (B)	P_i^{sc}	1 μW	TBC (Q3)
Quantum efficiency	η	0.8	TBC (Q3)
Charge range / stray floor	q	$[10^3, 10^8] e$	TBC (Q4)
Intracavity profile	G_i	uniform assumed	TBC (Q5)

Validation. The pipeline reproduces the published on-resonance strain sensitivities $h_{\text{min}} = 1.02 \times 10^{-22}$ (100 kHz) and 7.6×10^{-21} (10 kHz) at ratios 3.95 and 4.66 respectively — a *consistent* systematic offset compatible with the published damping table being angular rates ($\sqrt{2\pi} \approx 2.5$) times $O(1)$ bookkeeping; the 10-vs-100 kHz *scaling* agrees to 18% (Question 1). Four further internal validation tests (single-oscillator FDT peak, weight-independence of thermal S_h , per-mode SNR vs. full integral, integration-grid convergence to 10^{-4} of brute force) all pass.

3 Optimization setup

Annealed Metropolis walk minimizing $E = -\log d_{\text{max}}$: log-space multiplicative proposals on positive variables (clipped to bounds), tangent-space Gaussian on $\mathbf{w}$; geometric annealing $T : 0.2 \rightarrow$

0.002 (in $\log d_{\max}$ units); 10^5 steps per chain; three chains per case (one seeded at the as-built corner, two random); best-ever-visited state returned. Design variables and bounds: $f_{\text{trap},i} \in [10, 100]$ kHz; $q_i \in [10^3, 10^8] e$ (coupled cases); $Q_{\text{eff},i} \in [10^3, 10^8]$ (upper bound = engineering-plausibility placeholder, Q2); $\mathbf{w}$ free for readout B only. Three-way comparison: (i) *corner system* — as-built baseline: uniform traps at 100 kHz, charges at the stray floor, $Q_{\text{eff}} = 5.4 \times 10^5$, uniform weights; (ii) *uncoupled opt* — charges removed, everything else optimized (null hypothesis); (iii) *coupled opt* — everything optimized.

4 Results

Table 2: Three-way comparison, $d_{\max}$ at $\mathcal{M}_c = 10^{-3} M_{\odot}$ [AU]. v5: NA = 0.1, Q_{eff} fixed at 5.4×10^5 . v6: NA = 0.5, per-disc Q_{eff} optimized.

Case	Readout A		Readout B	
	v5	v6	v5	v6
Corner system	1,138,752	1,138,752	83.9	419.7
Uncoupled opt	1,138,752	1,138,752	208.9	2,260.6
Coupled opt	1,138,752	1,138,752	380.9	4,904.9
coupled/uncoupled	1.000	1.000	1.82	2.17

Readout A (thermal-limited in its bucket): exact three-way tie at $d_{\max} = 5.52$ pc. The optimum is the uniform 10-disc stack at the 100 kHz trap ceiling; its bucket bottom (1.3×10^{-22}) reaches the LSD target band. Coupling is provably harmless there (any coupling network exerts zero net force on the common mode of identical traps) and empirically unhelpful; Q_{eff} is a null direction (signal and thermal noise amplify identically).

Readout B (shot-limited): the hierarchy is strict and converged (seed spread 3–6%): corner < uncoupled opt < coupled opt. The v6 winner has traps in two clusters (20–32 and 74–100 kHz), charges $(0.3\text{--}1) \times 10^8 e$, modes spread 63–383 kHz — several *above* the trap ceiling, reachable only through coupling and fully driven under Eq. (5) — and every Q_{eff} pinned at $(0.6\text{--}1) \times 10^8$: the deepest buckets sit *on the thermal floor* inside the LSD target band (Fig. 1). Because Q_{eff} pins at the search bound, readout B’s absolute reach is *bound-limited*: it scales directly with the practical Q_{eff} cap (Question 2). The relative statements (coupled beats uncoupled 2.17×; both beat the as-built corner) are insensitive to global factors.

5 Questions for Nancy

Q1. Units of the damping table (factor 2π). In PRL 128 Table I, are $N_i \gamma_g$ (0.17/1.7 Hz) and γ_{sc} (0.05/0.005 Hz) plain frequencies (cycles/s) or angular rates (rad/s) labeled Hz? The FDT force noise needs the angular rate; the answer moves every absolute $\sqrt{S_h}$ and $d_{\max}$ by $\sqrt{2\pi} \approx 2.5$ ($16\times$ in event rate). Our residual factor ~ 4 against the published $h_{\min}$ is consistent with “angular” plus $O(1)$ bookkeeping.

Q2. Practical ceiling on Q_{eff} . How high can the feedback-cooled Q realistically go — i.e. how weak can cold damping be while (a) the trap frequency stays stable to within a linewidth over the $\sim 1/\Gamma$ coherence time (laser intensity noise), and (b) feedback measurement-noise

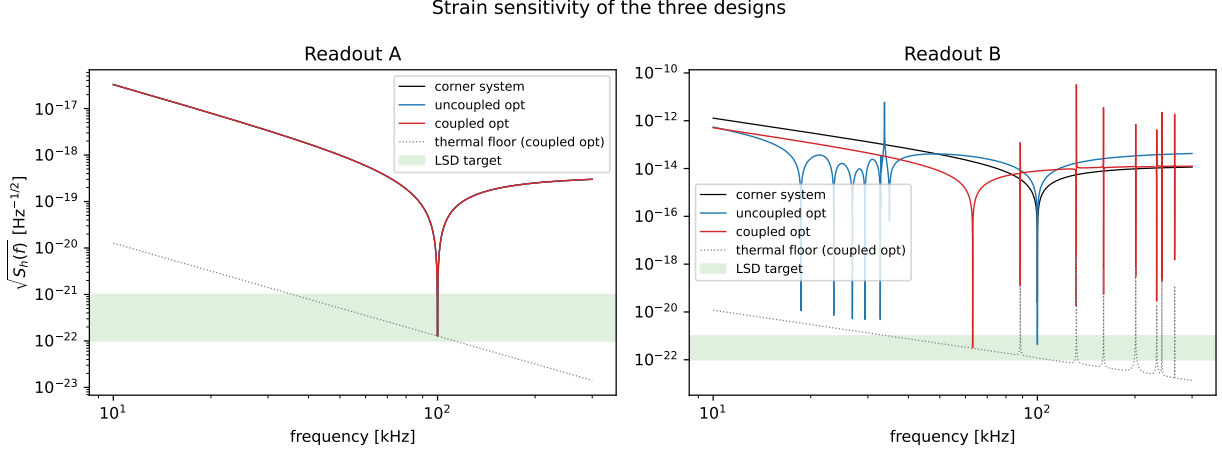


Figure 1: Strain sensitivity of the three designs (v6). Left: readout A (tie; single bucket at 100 kHz on its thermal floor). Right: readout B — the coupled optimum (red) tiles buckets across and beyond the trap range; its deepest dips touch the thermal floor (gray dotted) inside the LSD target band (green).

injection remains negligible? Our optimizer pins at the arbitrary search bound 10^8 ; readout B's forecast scales with the true cap.

Q3. Readout B (imaging) parameters. Is $NA = 0.5$ an operating point or an upper limit, and is it simultaneously achievable for all ten discs? Per-disc collected power P_i^{sc} (we assume $1 \mu\text{W}$), imaging wavelength (532 nm?), and detector quantum efficiency η (0.8?). These set the shot plateau, Eq. (11), and hence all readout-B numbers.

Q4. Charge range and stray floor. What charge can be controllably placed on a disc (upper bound), and what is the residual/stray charge floor (lower bound)? These bound the Coulomb design space, Eq. (2); the coupled optimum currently uses $q \sim 10^8 e$ — is that feasible on a $75 \mu\text{m}$ stack?

Q5. Intracavity coupling profile G_i . Are the ten discs' dispersive couplings equal (all at equivalent antinode positions), or is there a known non-uniform profile? A non-uniform hardwired profile is the one regime where coupling could also matter for readout A (mechanically re-aligning modes with the fixed weights).

Q6. Drive convention. The companion note's Eq. (20) ($\beta_i = k_{\text{trap},i}L$, from $\delta x = \hbar L/2$) vs. AG13's $F = \frac{1}{2}m\omega_{\text{gw}}^2\ell_m\hbar$: we have adopted AG13 (exact, full cavity length). Please confirm the intended L convention in the note (half- vs. full baseline) — a factor 2 in amplitude, 4 in power.

Q7. FDT temperature. We use the 300 K gas temperature in Eq. (8) with the gas γ_g (feedback treated as noiseless cold damping). Confirm this is the right bookkeeping for the planned operating mode (vs. an effective center-of-mass temperature).

Q8. Disc mass. Our geometric estimate $5.8 \times 10^{-10} \text{ kg}$ (SiO_2 spacer $14.58 \mu\text{m}$ + two 110 nm Si caps at $r = 75 \mu\text{m}$) matches the PRL 128 bookkeeping — confirm.

Planned next steps

Frequency-dependent weights $\mathbf{w}(f)$ for readout B (strictly optimal post-processing; removes anti-resonances); non-uniform G_i study (pending Q5); stray-charge/trap-disorder tolerance budget; detection-rate objective over the PBH mass function; and an axion-scan objective (monochromatic source at unknown frequency — the science case where band coverage, and possibly coupling-spread arrays, are natively favored).